

# **BINF 5503**

# **Data Analytics and Storytelling**

Week 6

Module 4 – Scientific Literacy

# Outline

- Ethics in research & research misconduct
- Literacy when interpreting scientific claims
- Bad science and bad storytelling
- The review process

# What is research misconduct?

- Fabrication or falsification of data
- Plagiarism
- Financial misconduct

# Fabrication vs. Falsification

| Falsification                                                                                                                                                                                                                                                                      | Fabrication                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>▶ Manipulating data or the experimental procedure to produce a desired outcome</li><li>▶ Throwing away unwanted results</li><li>▶ A way of avoiding complicated or inexplicable results, make data fit the hypothesis better, etc.</li></ul> | <ul style="list-style-type: none"><li>▶ Recording or presenting fake data</li><li>▶ Completely making up results</li></ul> |

# Fabrication vs. Falsification

You want to show that mycorrhizal fungi in soil benefits plant growth. You set up a greenhouse experiment with plants that have mycorrhizae in the soil and those in sterilized soil.

Unfortunately, you don't see the results you were looking for – any differences in plant size are insignificant. So, you shift all the treatment plants over to a sunnier area to try and boost growth.

Falsification



# Fabrication vs. Falsification

## **Fabrication**

After two months sampling purple loosestrife, you return to the lab with your data set. You then realize you forgot to measure plant height for one of the populations you sampled. You don't want to have missing data, so you quickly fill in that column with plant heights, based on the other measurements you took at nearby locations.

# Fabrication vs. Falsification

You hypothesize that herbivory is greater on native versus non-native trees. You collect leaves from across your sampling region. In the lab, you quantify percent area damage, but you notice that many of the native leaves actually have *less* herbivory. In an attempt to salvage your hypothesis, you decide to “subsample”

– i.e., you focus all your attention on the leaves that show significant damage.

**Falsification**



# Plagiarism

- Using someone else's words, ideas, or results without attribution
  - Can be intentional or unintentional (through incorrect referencing)
- Replicating substantial elements of your own published work without acknowledgement

# Plagiarism

- An author submits a paper for review in which one of the figures used was previously published in an article of his own works.

**Is this wrong, and if so, what should be done?**

## Self-Plagiarism

Solution: repeat the experiment or part of an experiment and include new data

**OR a more reasonable solution:** cite the original paper!

# Financial misconduct

- Misuse of funds acquired for support of research
  - Failing to comply with the terms & conditions of grants or contracts
  - Failing to identify the source of research funds
- Funding is important for conducting research, but easy to misuse

# Misleading publications

- Fraud
- Publication bias
- Retractions

# Fraud

- Deliberate misconduct
- Most fraud occurs in the medical sciences. Why?
  - Incentive – more money in medical research
- In other fields, it is harder to catch fraud
  - e.g., in ecology & environmental sciences, even if the experiment is repeated, it's hard to get the same results

# Publication bias

- Greater likelihood of publication if you have significant results
  - Studies published in peer-refereed journals are much more likely to report statistically significant results than are studies that report a nonsignificant conclusion
- Interesting TED talk by Ben Goldacre about publication bias in clinical trials:
  - [What doctors don't know about the drugs they prescribe](#)

# Retractions

- When a published scientific paper is withdrawn due to experimental errors or scientific misconduct.
  - A way of correcting the scientific record
- Retractions and plagiarism cases have risen steadily over the past decade. Why?
  - Software detecting plagiarism
  - More people reading papers thanks to easy access on Internet



# Data forensics

- Regular data exploration (looking at data spread, looking for outliers, etc.) is insufficient if data is faked
  - Look for duplicated sequences, too-precise relationships, too-good results
- R packages developed specifically used to look for falsified data
  - ``digitTests``
  - [Machine Learning methods](#)

# Research Ethics

- Impossible to be constantly checking collaborator's data
  - Use judgement, validate, replicate
- Scientific misconduct is unethical and unacceptable in science
- Keep raw data, use open data, publish code
- Scientific misconduct does not just harm researcher, but also collaborators and the field
- Publish or perish mindset may be costly to scientific truth

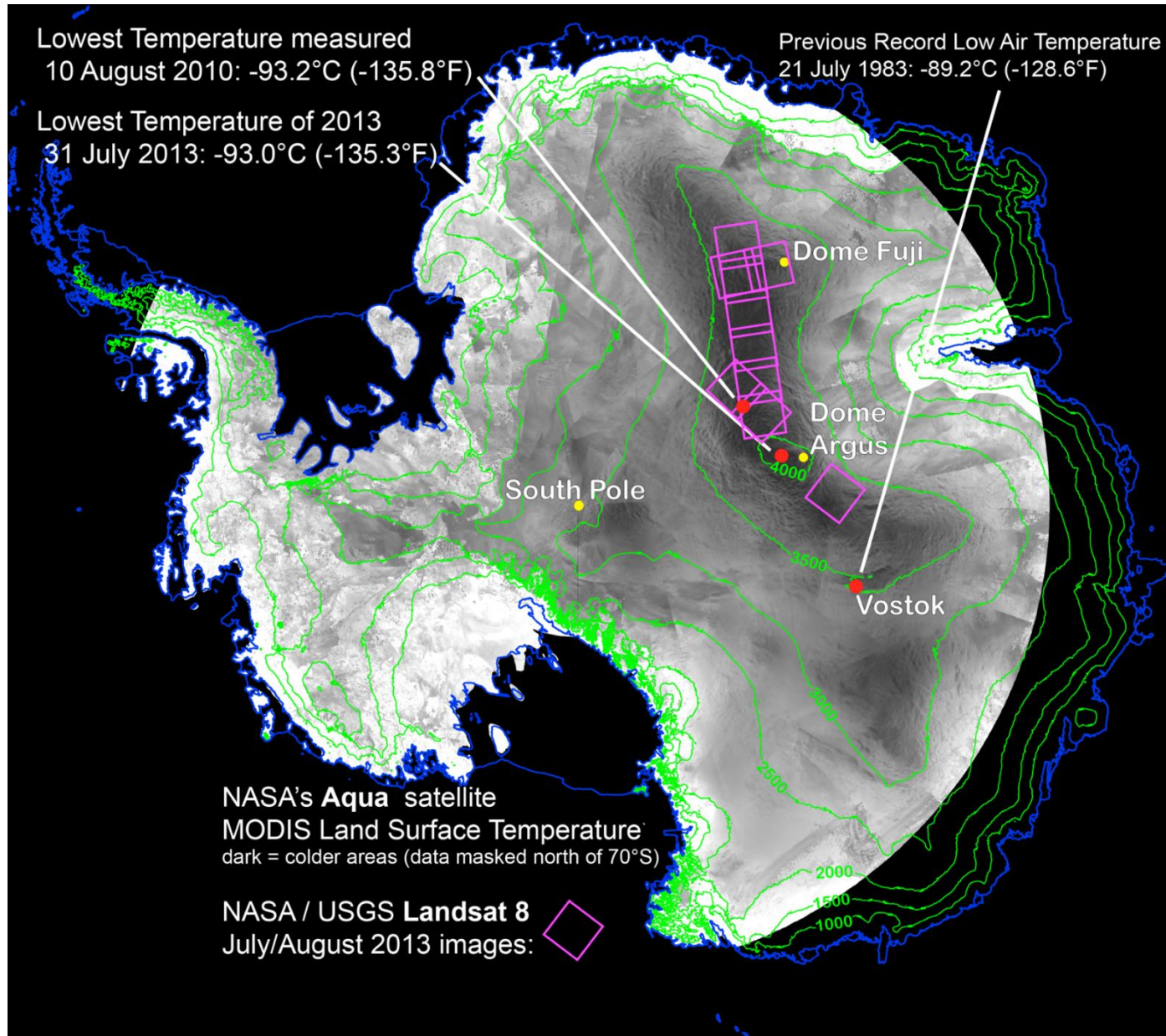
# Scientific Literacy

Recommended reading:

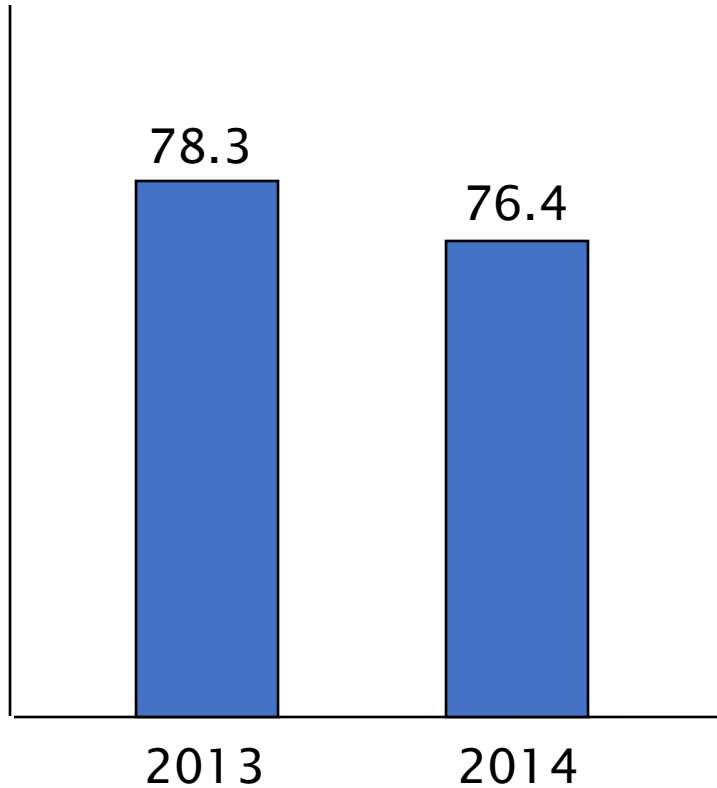
[Twenty tips for interpreting scientific claims](#) (Sutherland et al., 2013, Nature)



# 1. Differences and chance cause variation

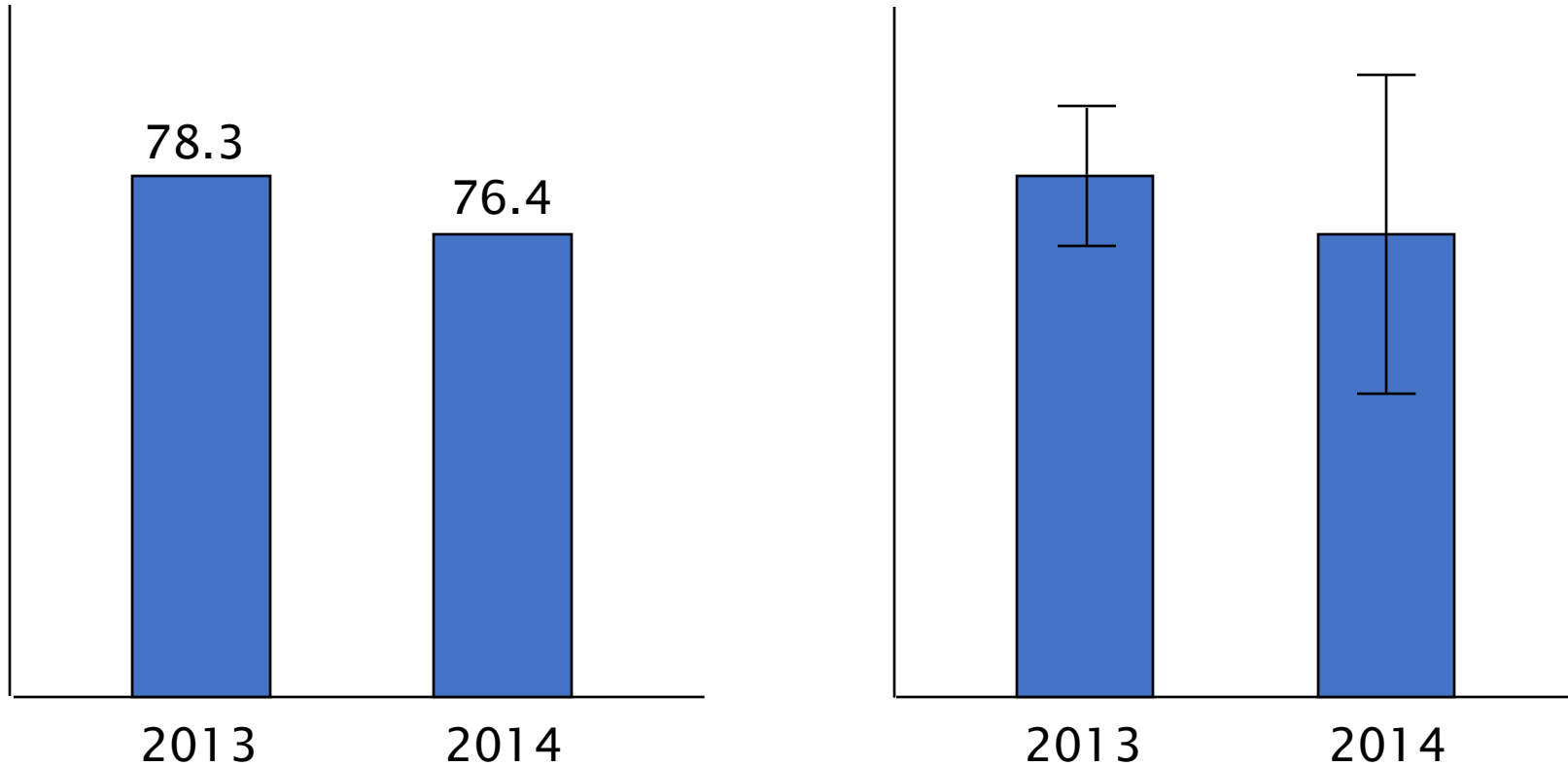


## 2. No measurement is exact



Average Life Expectancy of Canadian Men

## 2. No measurement is exact



Average Life Expectancy of Canadian Men

# 3. Bias is rife

Sources of bias:

- Experimental design
- Researchers try to get significant results to increase likelihood of publishing
- Researchers influenced by the fact that they know which treatments were used for each sample
- Participants receiving a treatment might act differently if they know what the researcher is looking for

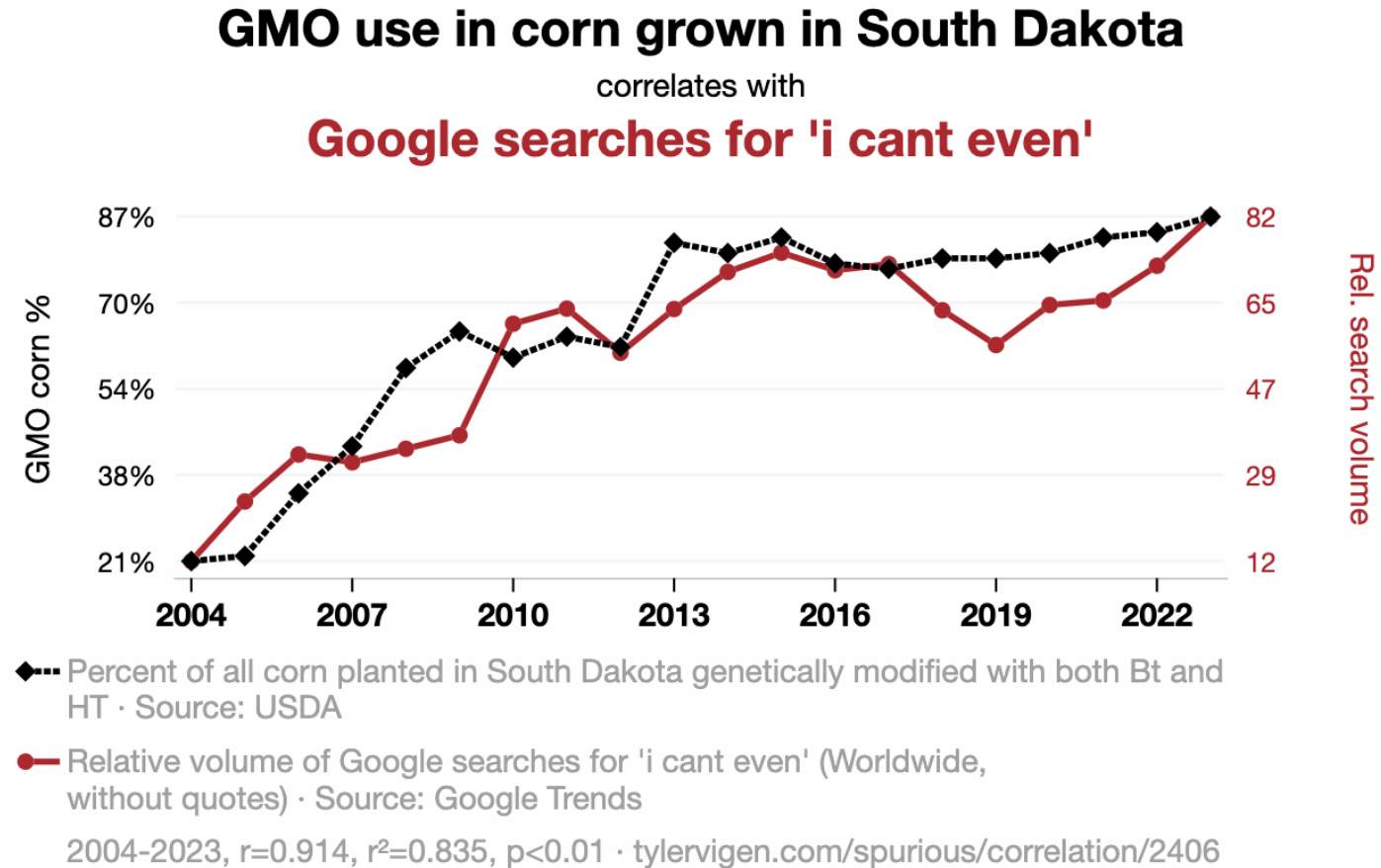
**Important terminology:**

- Double-blind study
- Confirmation bias

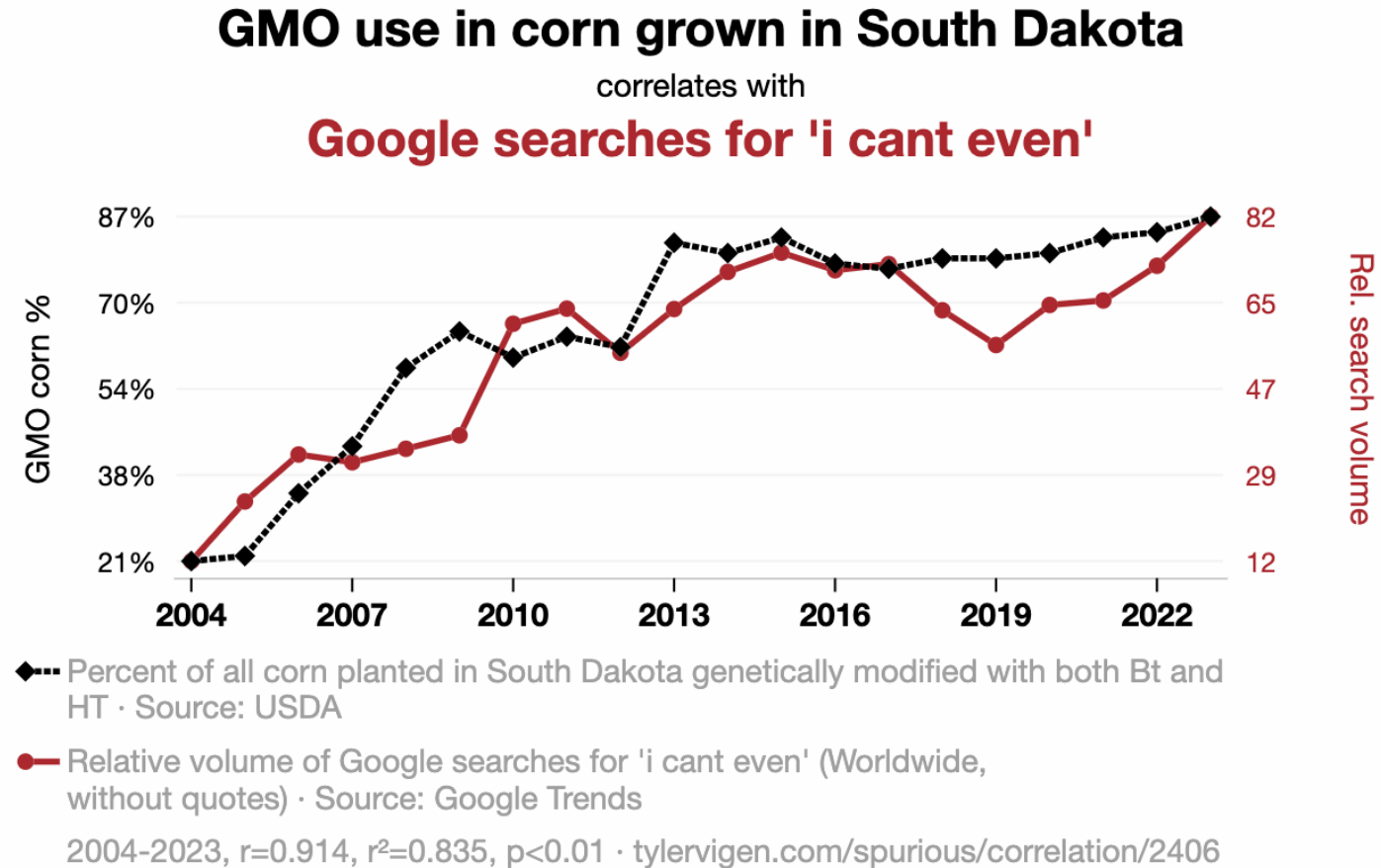
## **4. Bigger is usually better for sample size**

- The more observations, the more variation can be taken into account
- Estimates become more reliable
- Amount of error is reduced

# 5. Correlation does not imply causation

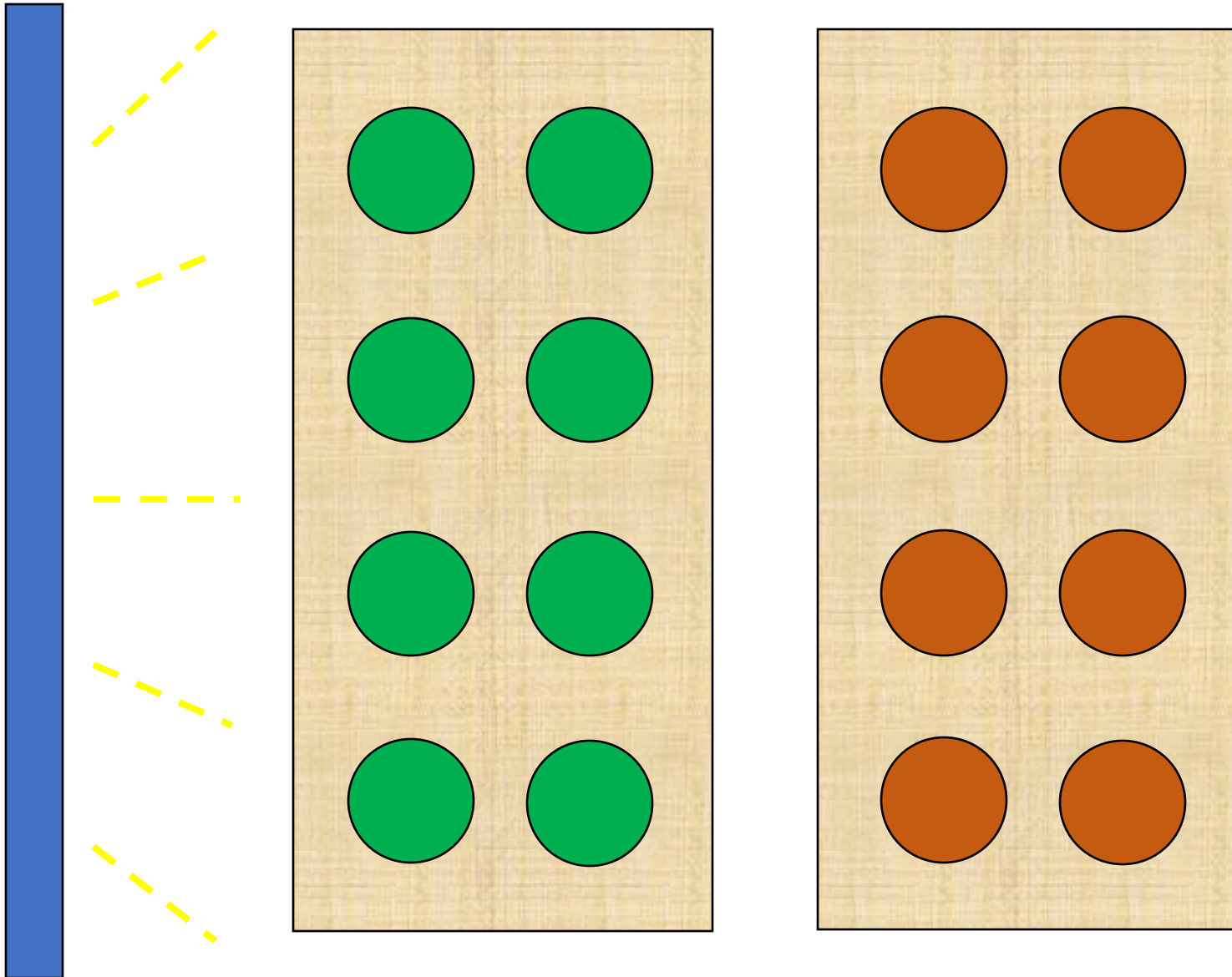


# 5. Correlation does not imply causation

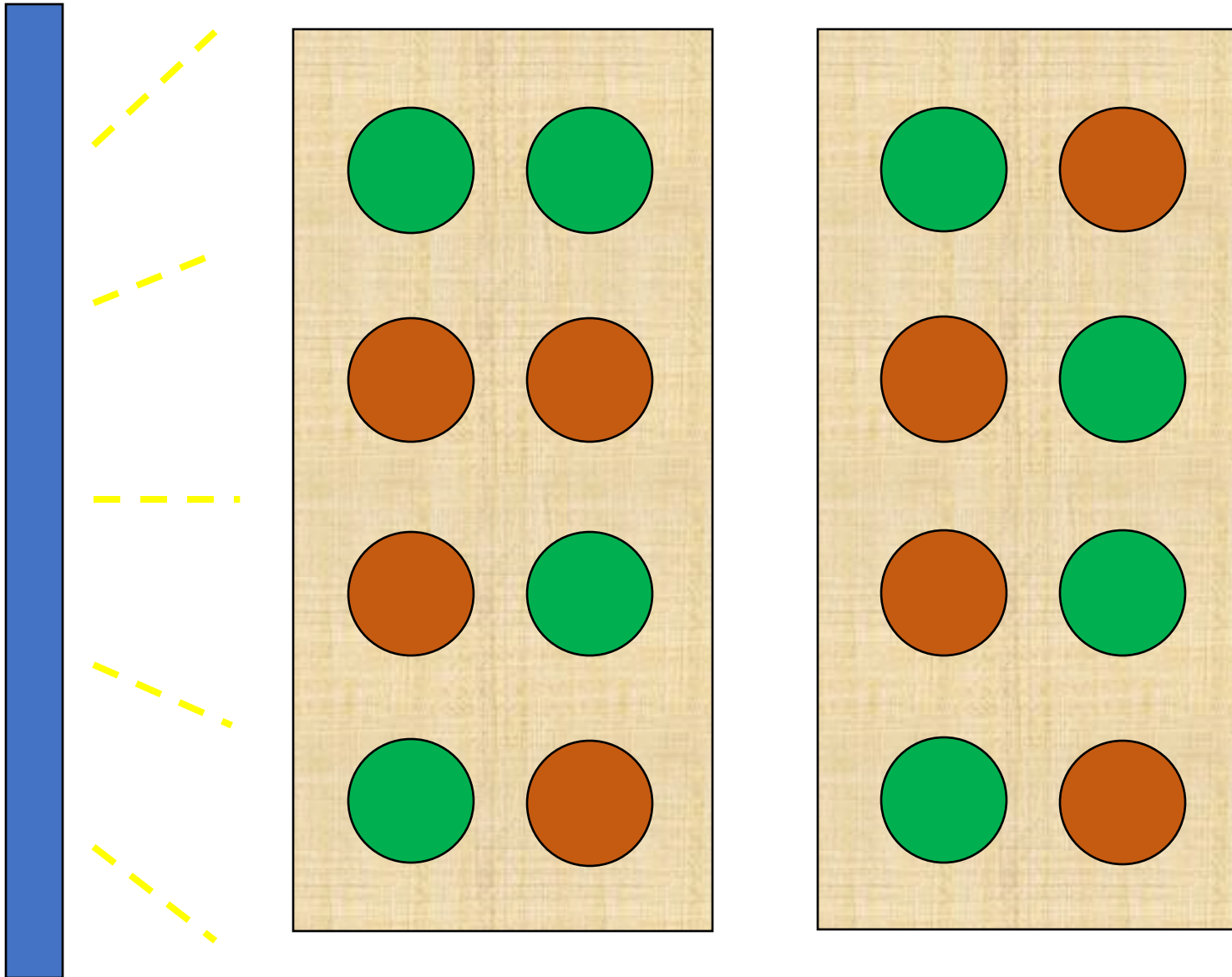


- Coincidence
- Lurking variable (aka confounding factor)
- Causation

# 10. Randomization avoids bias



# 10. Randomization avoids bias



# 11. Seek replication, not pseudoreplication

When replicates/samples in an experiment are not statistically independent

Example:

- A researcher wants to determine the effect of two different teaching methods on the performance of university students. A class of 500 students is split into two lecture groups of 250 students each. One group is taught through lectures and the other is taught through discussion-based learning.
- *What is the sample size in this experiment?*

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# 19. Data can be dredged or cherry-picked

**Dredged:** found a correlation in a massive data set

- Can almost always find some correlation between two variables

**Cherry picked:** scientists only report some variables/factors/results to support their hypothesis

- In both cases you have to ask yourself ***what am I not being told?***

# Bad Science & Bad Storytelling



# Anecdotes are not evidence

- Isolated examples of interesting stories can be misleading
- Unpublished studies that have not been assessed may be wrong
- Science is based on scrutiny
- Important to publish negative or non-significant results

# Beware of “breakthroughs”

- Single studies rarely resolve an issue
- Single studies can get it wrong!
- Look out for sample sizes, number of participants, rigorous statistical methods
- Media coverage is biased
  - There are other journals apart from Nature and Science!
  - May not reveal full paper or present the full story
  - Stories may take on increased significance (relevance to your own personal life)

# Science is not anonymous

- Make sure you can find the original publication
- “Researchers at Humber College...” – who? Which lab?

# Science changes to match evidence

- Science is self-correcting
  - Gaps & contradictions in scientific evidence always used to discredit science
  - Goal: identify gaps, resolve contradictions
- Not a conspiracy when new evidence emerges and results change
  - Demand for immediate answers & failure to provide answer  $\neq$  proof of conspiracy
  - Need time to gather objective evidence that is not speculative or anecdotal
- Critically evaluating research is key
- Pseudoscience generally does not change to match new evidence

# Make sure your storytelling is scientifically accurate

- Research must include fair tests of hypotheses
- Report research carefully and completely
- All authors have direct contribution to research
  - Equal contribution or order reflects magnitude of contribution
  - [Authorship contribution statement](#):

“VMZ and SG collected and cleaned the data, YHP and VMZ conducted the analyses equally, YHP and SG wrote the first draft, all authors contributed to the writing of the manuscript and creation of the poster.”

# Make sure your storytelling is scientifically accurate

- Be careful of terminology you use

- Some terms are better for scientific writing (e.g., on poster), and some are better suited for oral presentations

- Significance should be defined

- E.g.,  $p = 0.043$
  - $x$  “significantly affects”  $y$  is different from “trends” – the result is *because* of the cause (not by chance)
  - Avoid “prove”; instead, results “support”

- Connect results to real-world implications carefully

- Put results into context
  - Do not exaggerate or overstate

## Terms that have different meanings for scientists and the public

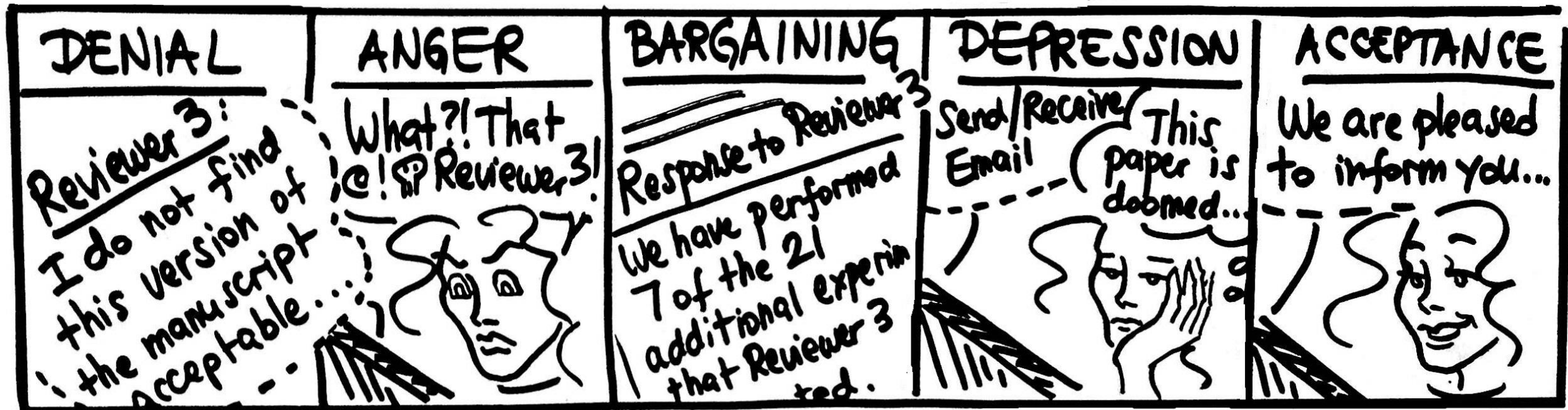
| Scientific term   | Public meaning                | Better choice                         |
|-------------------|-------------------------------|---------------------------------------|
| enhance           | improve                       | intensify, increase                   |
| aerosol           | spray can                     | tiny atmospheric particle             |
| positive trend    | good trend                    | upward trend                          |
| positive feedback | good response, praise         | vicious cycle, self-reinforcing cycle |
| theory            | hunch, speculation            | scientific understanding              |
| uncertainty       | ignorance                     | range                                 |
| error             | mistake, wrong, incorrect     | difference from exact true number     |
| bias              | distortion, political motive  | offset from an observation            |
| sign              | indication, astrological sign | plus or minus sign                    |
| values            | ethics, monetary value        | numbers, quantity                     |
| manipulation      | illicit tampering             | scientific data processing            |
| scheme            | devious plot                  | systematic plan                       |
| anomaly           | abnormal occurrence           | change from long-term average         |

# Make sure your storytelling is scientifically accurate

- Consensus with collaborators and scientists
  - Base conclusions on empirical data
- Science is ongoing
  - Gaps in research & different results warrants further research
- Transparency in review process
  - Funding sources
  - Original datasets
- Review your research before presenting

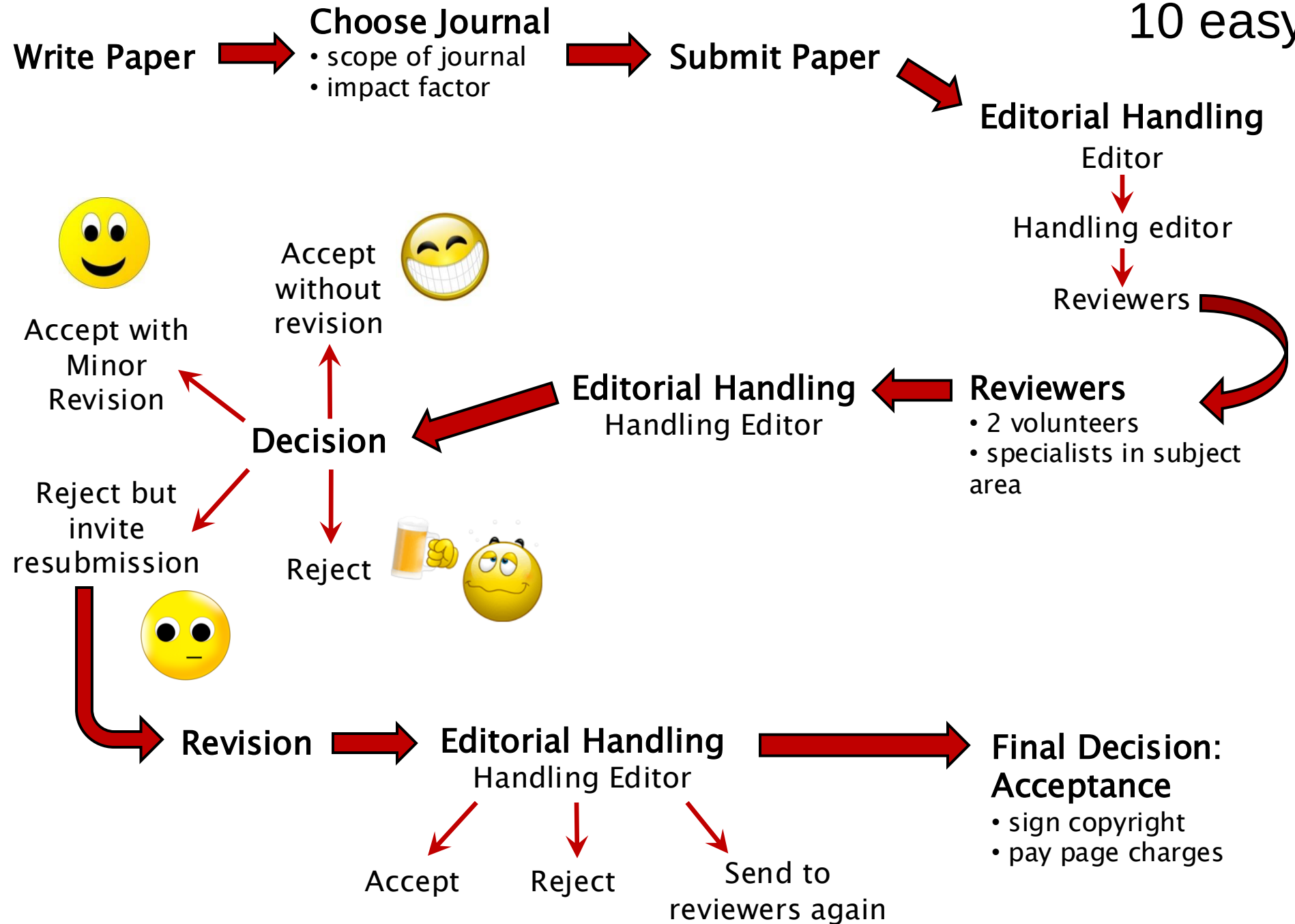
# The Review Process

## THE FIVE STAGES OF PEER REVIEW



XYKADENIQ2 2014

# 10 easy steps



# The review process

- Typically relies on 2 reviewers and 1 editor
- Takes 6-12 months, if not longer
- Authors don't get paid
  - Neither do reviewers or handling editor
  - Process often costs you money
- Most papers are rejected
  - >80% rejection rate for major journals

# Impact factors

- Average number of citations the articles in a journal receive
- Used as a proxy for the relative importance of a journal within its field
- Calculation:
  - Average number of citations received per paper published in that journal during the **two preceding years**
  - E.g., if a journal has an impact factor of 3 in 2015, then papers published from 2013-2014 received an average of 3 citations each

# Criticism of impact factors

1. Highly discipline-dependent
2. Impact factor is based on the **mean** number of citations per paper, which can be misleading
3. People tend to judge the quality of a paper based on the impact factor of the journal it was published in
4. A journal can adopt editorial policies that increase its impact factor

# Why are papers rejected?

- Manuscript does not fall within journal's aim/scope
- Bad writing
- Manuscript does not match style or guidelines of the journal
- Research does not add value
- Lack of evidence or insignificant results
- Unclear hypothesis
- Unclear methods or analysis
- Violation of research ethics

# Reviewers' comments and authors' responses

- Language is professional but not unkind
- Responses should not be defensive/combatative nor groveling/pandering
  - Authors can agree or disagree, but it should be justified
- What if the reviewers' comments do not agree?
  - Handling editor will recommend that authors largely focus to one reviewer's comments
- **Satisfy the editor that you have addressed all the reviewers' concerns**

# Peer Review Blueprint

- Decline if you don't have the expertise of time
  - Be honest with affiliations to avoid a conflict-of-interest
- Start with a summary of the paper
- Substantiate critique with evidence in the submitted paper/poster
- Do not go overboard with criticisms
- Justify requests for more work
- Avoid contradictions

Reviewing is a time-sink but it is crucial for the scientific community, and also for yourself as a scientist

# Predatory journal

- “write-only publishing” or “deceptive publishing”
  - Lack of a definition – what makes a journal “predatory”?
- Exploitative
- Charges publication fees without checking for quality or legitimacy
- Articles are not widely circulated
- Authors receive no citations and lose rights over their paper
- Predatory conferences now increasingly popular:
  - pretend to be legitimate scientific conferences, but their purpose is to profit from registration fees without any qualitative contribution to the scientific community

Credible journals



Peer review, Revisions, Rejections

Predatory journals



Greetings, We Adore Your Research!

# Recap

- Abide by scientific ethics in research, and avoid research misconduct
  - Scientists have a duty to avoid bad science and bad storytelling
  - When interpreting and/or reporting scientific claims, make sure that the science is sound!
  - The review process is important for the scientific field, but be aware of predatory journals and conferences